

POINTER BASICS — C Programming

SLIDE 1 — What is a Pointer?

Definition:

A **pointer** is a variable that stores the **memory address** of another variable.

Simple idea

Variable → **Value**

Pointer → **Address of variable**

Example

```
int x = 10;
int *p = &x;

printf("Value = %d\n", x);
printf("Address = %p\n", (void*)p);
```

Explanation

- `x` stores `10`.
- `&x` gives the **address** of `x`.
- `p` stores that address.
- `*p` can access the value stored at that address.

Remember:

Pointer = stores an address.

SLIDE 2 — Pointer Declaration

Structure

```
data_type *pointer_name;
```

Example

```
int x = 25;
```

```
int *p = &x;  
  
printf("%d", *p);
```

Output

25

What happens?

```
x = 25  
↓  
&p = address of x  
↓  
p stores address  
↓  
*p gives 25
```



Remember:

& → **address of**

***** → **value at address**

SLIDE 3 — Accessing Value Using Pointer

The ***** operator is called the **dereference operator**.

Example

```
int x = 50;  
int *p = &x;  
  
printf("Value: %d\n", *p);  
  
*p = 100;  
  
printf("New value: %d", x);
```

Output

Value: 50
New value: 100

Why?

*p refers to the value stored at the address held by p.

⚡ Changing *p changes x.

SLIDE 4 — Pointer and Variable

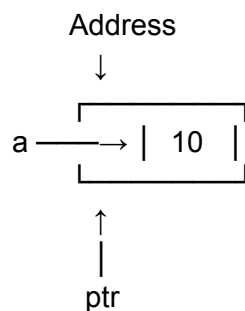
Example

```
int a = 10;  
int *ptr = &a;  
  
printf("a = %d\n", a);  
printf("*ptr = %d\n", *ptr);
```

Output

```
a = 10  
*ptr = 10
```

Quick View



🔍 A pointer does **not** normally store the value directly; it stores the **address** where the value is located.

SLIDE 5 — Pointer Basics Recap

Flow

Declare variable



Get address using &



Store address in pointer



Access value using *

5 Things to Remember

1. Pointer stores a **memory address**.
2. **&** gives the address.
3. ***** accesses the value at an address.
4. Pointer type should match the pointed-to data type.
5. Always initialize pointers before using them.

Final Example

```
int n = 20;  
int *p = &n;
```

```
printf("%d", *p);
```

Output:

20



Memory Trick:

& = Address → *** = Value**



QUESTIONS

Q1. MCQ

What does a pointer store?

- A) Character
- B) Memory address
- C) Keyword
- D) Operator



Hint: A pointer tells us where a variable is stored.



Glossary: Address → location of data in memory.

✓ Answer: B) Memory address

Q2. What is the output?

```
int x = 30;  
int *p = &x;  
  
printf("%d", *p);
```

- A) Address of x
- B) 0
- C) 30
- D) Error

💡 **Hint:** *p accesses the value.

📖 **Glossary:** Dereference → access the value through a pointer.

✓ Answer: C) 30

Q3. Fill in the blank

```
int x = 10;  
int *p = _____;
```

💡 **Hint:** Use the operator that gives a variable's address.

📖 **Glossary:** & → address-of operator.

✓ Answer: &x

Q4. Which operator is used for dereferencing?

- A) &
- B) *
- C) %
- D) #

💡 **Hint:** It is also used when declaring a pointer.

 **Glossary:** Dereference → get the value stored at an address.

 **Answer: B) ***

Q5. What happens here?

```
int x = 5;  
int *p = &x;
```

```
*p = 15;
```

- A) **p** becomes 15
- B) **x** becomes 15
- C) **x** becomes 5
- D) Error

 **Hint:** ***p** refers to **x**.

 **Glossary:** Assignment → giving a value to a variable.

 **Answer: B) x becomes 15**

Q6. Arrange the steps

- A. Store address in pointer
- B. Declare a variable
- C. Access value using *****
- D. Get address using **&**

 **Hint:** Start with the variable.

 **Glossary:** Pointer → variable that stores an address.

 **Answer: B → D → A → C**